

LECTURE 37: MATH1061/7861
TOPICS: MATRIX REPRESENTATIONS OF
GRAPHS
DATE: TUESDAY MAY 26, 2026

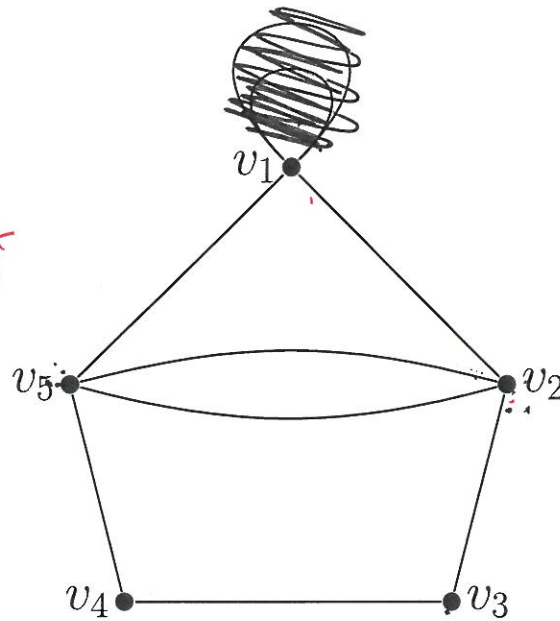
Learning objectives:

- (1) Represent a graph by an adjacency matrix.
- (2) Draw a graph from its adjacency matrix.

** Double check your grades on GradeScope
and Blackboard*

Note: Matrix multiplication is NOT assessable

Q1: Write down the adjacency matrix for the following graph.



Square matrix

Same
number
of rows
& columns.

entries
should be
non-~~neg~~ negative
integers.

Diagonal.

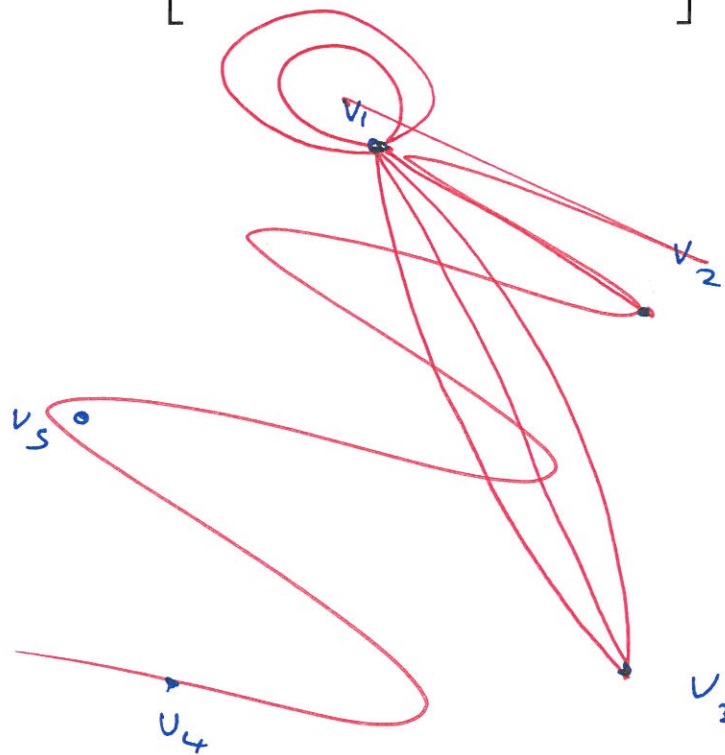
	v_1	v_2	v_3	v_4	v_5
v_1	0	1	0	0	1
v_2	1	0	1	0	2
v_3	0	1	0	1	0
v_4	0	0	1	0	1
v_5	1	2	0	1	0

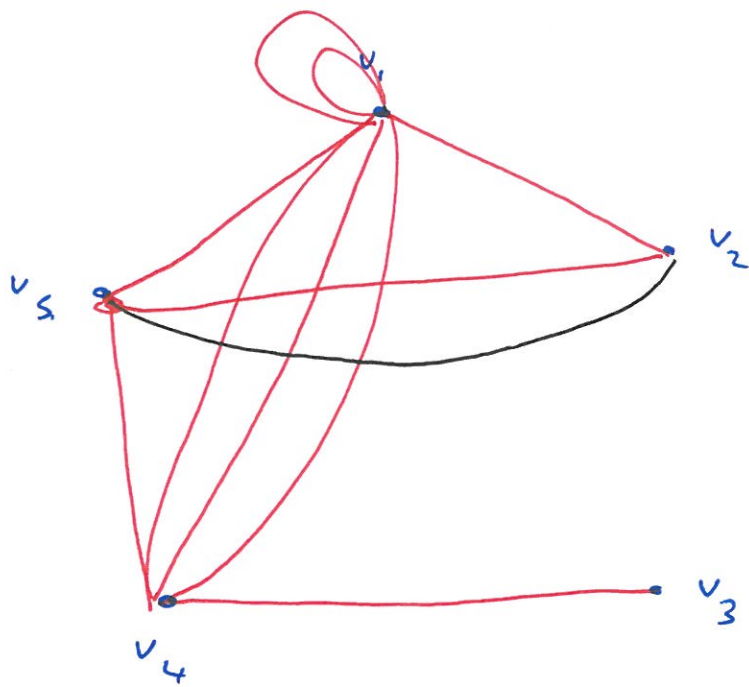
5x5

numbers are recording walks
of length 1.

Q2: Draw the graph associated to the adjacency matrix

$$\begin{bmatrix} 2 & 1 & 0 & 3 & 1 \\ 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 3 & 0 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 \end{bmatrix}$$





$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 \end{bmatrix}
 \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 1 & 1 & 2 \\ 2 & 6 & 0 & 0 & 0 \end{bmatrix}$$

What do these entries mean
in terms of the graph?

Records walks of length 2 in our graph.

Theorem: Let A be the adjacency matrix of a graph. Then for each positive integer k , the matrix $A^k = \underbrace{A \times \dots \times A}_{k \text{ - times}}$ records the number of walks of length k in our graph.

Challenge: How to tell whether a graph ~~is~~ is connected just from the adjacency matrix?